

OPENHOARD · IT OVERVIEW

The AI filesystem

Remembers everything. Guards it all.

One governed index over SharePoint, file shares and cloud buckets, so people and their AI assistants find files by meaning, with permissions enforced on every request.



Open source (Apache-2.0) · Self-hosted · Pre-alpha · September 2026

Folders were built for filing cabinets

54%

of US office workers waste time searching for files in disorganized systems

Source: Elastic / Wakefield Research

77%

of organizations had a Microsoft 365 governance incident in the past year

Source: ShareGate survey

38%

had former employees or guests keep access they should have lost

Source: ShareGate survey



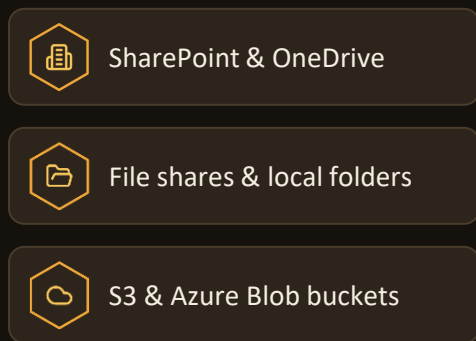
The AI multiplier

AI assistants read everything a user can access. Oversharing that used to be hidden is now one prompt away, so security teams block AI instead.

WHAT IT IS

One governed layer over your existing storage

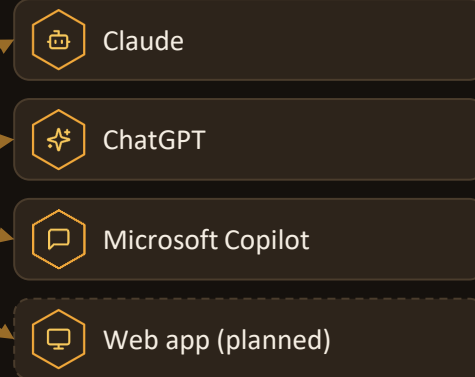
WHERE FILES LIVE



OpenHoard

index · tags · policy · search · audit

HOW PEOPLE REACH THEM



Leave files in place

Index SharePoint and shares where they are. Move files into buckets only when you're ready.

Your identity, your rules

Entra ID / OIDC sign-in, SCIM groups, and access that follows tags.

Bring any AI

No custom model or agent. Any assistant that speaks MCP connects as the signed-in person.

Architecture at a glance

SOURCES

SharePoint
OneDrive too · next up

File folders
local & network · today

S3 · Azure
blob storage

INGEST

Connectors
SDK v1: crawl, delta, ACL import

Sync runner
schedules, checkpoints, mass-delete guard

TRUSTED CORE

Catalog
PostgreSQL + row-level security

Blob store
BLAKE3-hashed, tenant keys

Enrichment
extract, summarize, tag

Policy
Cedar rules, fail closed

Search
full-text + pgvector

Audit
hash-chained log

ACCESS

MCP server
OAuth 2.1, approved AI clients

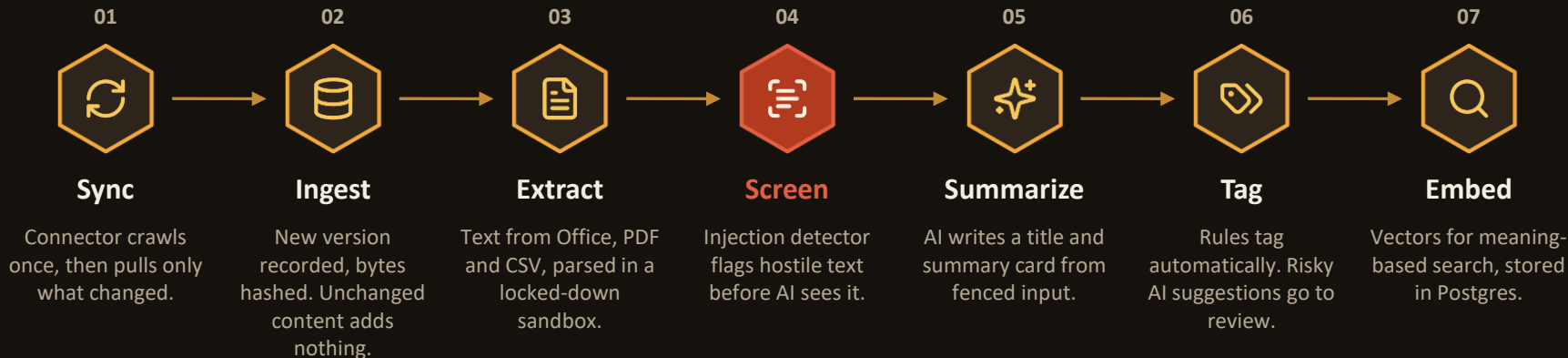
Admin API + CLI
clients, admins, sources

Web app
planned



Identity Entra ID or any OIDC sign-in · SCIM 2.0 users and groups · service accounts with scoped API keys

The ingest pipeline: from file to findable



Fail closed

A new or changed version stays hidden, and metadata-only for AI, until every step has finished.



Runs as background jobs

Postgres-backed queue: one job per version, retries with backoff, dead letters with a safe salvage path.

Where content, metadata and tags live



PostgreSQL

the catalog

- Files, versions, source pointers
- Tags, facets and grants
- AI cards + extracted text
- Embeddings (pgvector)
- Activity and audit log

Row-level security per tenant



Blob store

the bytes

- S3 (default), Azure Blob or local disk
- Keyed by BLAKE3 content hash
- Keys scoped per tenant
- Used when OpenHoard manages or copies a file

Identical content stored once



The original source

left in place

- Indexed files stay in SharePoint or the share
- OpenHoard keeps a pointer, hash and metadata
- Re-reads only what changed
- Never moves a file without asking

No migration required

ZONES

Managed

stored in OpenHoard's buckets

Indexed

stays at the source

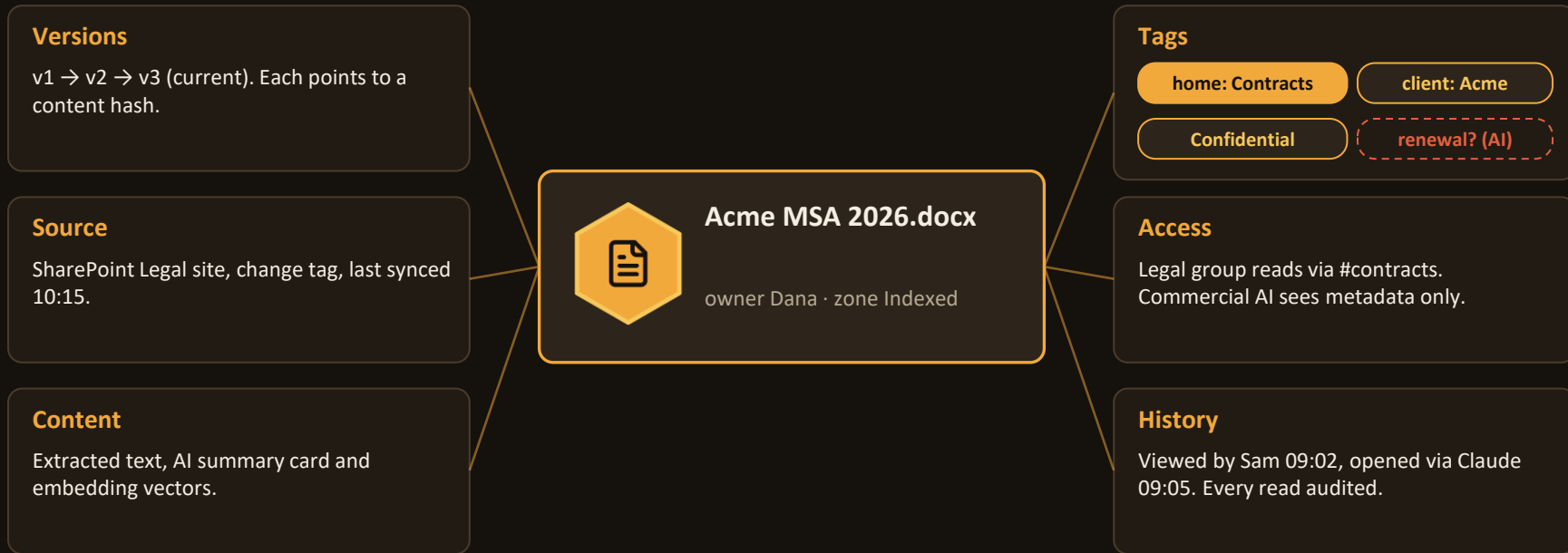
Local-only

never leaves the device (M2)

Code

source code repositories

Anatomy of one file



Illustrative example. A folder becomes just one more tag, so the same file can live in several 'places' at once.

Tags replace folders, and carry the permissions



Many tags, one home

A file can sit in Contracts, Acme and Q3 at once. One primary tag is its home.



People, rules and AI all tag

Rules tag on ingest. AI suggestions stay untrusted until a person approves them.



Grant access to tags, not files

'Legal can read #contracts' covers every current and future contract.



Untrusted tags only tighten

An AI guess can make a file more private, never less.

VISIBILITY LEVELS (PER TAG)

Readable open and read the content

Discoverable sees it exists, title only

Hidden not in search at all

AI EXPOSURE

Set per tag and per AI client trust. Example:
Confidential files reach a commercial AI as metadata only, never content.

Permission-aware search, zero AI tokens per query



1M

file versions in the vector search benchmark

312 ms

worst-case p95 query time at 1M

0

LLM tokens spent per search

Benchmark: PostgreSQL 18.3 with pgvector 0.8.1. Query embeddings run locally by default.

Built to survive the security review



Tenant isolation

Postgres row-level security, forced even for the app's own account.



Fail-closed policy

Cedar policy engine. Any error or ambiguity means deny.



Instant offboarding

Disable in Entra and the next request is refused (measured 3–60 ms).



Approved AI only

OAuth 2.1. An admin approves each AI client and its trust level.



Injection defense

48 of 50 attack payloads flagged, 0 of 20 benign files.



Sandboxed parsing

Parsers run in a locked-down process: read-only files, no add-ons, capped memory and time.



Tamper-evident audit

Hash-chained log of sign-ins, admin changes and every AI read.



Text never authorizes

A document can inform an agent, but can't grant access or trigger actions.

What people see: they just ask their AI

Catch me up on the Acme renewal. What changed this week?

→ `openhoard.recent` → `openhoard.find`

Two files changed:

- Acme MSA 2026: Dana edited the renewal term on Tuesday.
- Acme pricing.xlsx: Sam added Q4 rates.



Acme MSA 2026.docx

Contracts · Acme · v3 · edited Tue

Open ↗

Illustrative. The assistant only sees what this person can see.

MCP TOOLS

find search

recent what changed

describe summary card

open link to open

tag suggest tags

explain why can I see it

SKILLS

find-and-open

catch-me-up

who-can-see-this

Each assistant connects as the signed-in person, with its own admin approval and token budget.

Fits your stack: no Docker, no black box



Identity and admin

- Entra ID or any OIDC single sign-on
- SCIM 2.0 users and groups
- Admins granted through an IdP group
- Service accounts with scoped API keys
- Admin CLI today, web console next



Runs on

- Windows, macOS or Linux
- Node 24 + PostgreSQL 17+ with pgvector
- Single-box mode with embedded Postgres
- Storage: S3, Azure Blob or local disk
- AI: Claude, OpenAI-compatible or local Ollama, with daily token budgets

TypeScript

Node 24

PostgreSQL

pgvector

pg-boss

Cedar

MCP

Apache-2.0

Where it stands: September 2026

48

pull requests merged

58

database migrations

~83k

lines of TypeScript, about half tests

3 OSes

CI on Windows, macOS and Linux

DONE



Single sign-on, SCIM, instant offboarding



OAuth 2.1 + MCP server for AI clients



Catalog, tags, policy engine, audit log



Text extraction + AI summaries + injection screen



Hybrid permission-aware search



MCP tools and skills



Scheduled sync of local folders

NEEDS A LIVE TEST

Sign-in against a real Entra tenant, and real AI clients over a public URL.

What's next



Order set by the maintainer: usable through Claude first, then SharePoint, then the admin console.

Stop filing. Start finding.

Keep your storage. Lose the chaos.



Pilot idea: one team share, one AI assistant



Tell us what your security review would ask



github.com/openhoard/openhoard

